ECA13 – MICROWAVE ENGINEERING

Experiment 1: Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

1. Aim

To simulate and analyze TEM, TE and TM propagation modes in microwave transmission structures using MATLAB, visualize their electric and magnetic field distributions, calculate cutoff frequencies and propagation constants, and compare their propagation characteristics.

2. Objectives

1. To develop simulation models for TEM, TE and TM propagation modes in suitable microwave transmission structures.
2. To visualize and analyze the electric-field and magnetic-field distributions associated with different propagation modes.
3. To determine the cutoff frequency and propagation constant of selected TE and TM modes.
4. To compare the propagation behavior of different modes and identify their suitability for practical microwave transmission.

3. Software Required

MATLAB

4.Program Coding

```
clc;

clear;

close all;

% TEM, TE and TM Propagation Modes

c = 3e8;

f = 10e9;

a = 22.86e-3;

b = 10.16e-3;

% Cutoff Frequencies

fc_TE10 = c/(2*a);

fc_TM11 = (c/2)*sqrt((1/a)^2+(1/b)^2);

% Propagation Constant
```

```
k0 = 2*pi*f/c;
beta_TE10 = k0*sqrt(1-(fc_TE10/f)^2);
% Field Grid
x = linspace(0,a,100);
y = linspace(0,b,60);
[X,Y] = meshgrid(x,y);
% Normalized Field Distributions
E_TE10 = sin(pi*X/a);
H_TE10 = cos(pi*Y/b).^2;
E_TM11 = sin(pi*X/a).*sin(pi*Y/b);
H_TM11 = abs(cos(pi*X/a).*cos(pi*Y/b));
% Create Figure
figure('Color','w','Position',[100 50 1400 700]);
% 1. TE10 Electric Field
subplot(2,3,1);
imagesc(x*1000,y*1000,E_TE10);
axis xy;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
title('TE10 E-field');
% 2. TE10 Magnetic Field
subplot(2,3,2);
imagesc(x*1000,y*1000,H_TE10);
axis xy;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
title('TE10 H-field');
% 3. TM11 Electric Field
subplot(2,3,3);
imagesc(x*1000,y*1000,E_TM11);
```

```
axis xy;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
title('TM11 E-field');

% 4. TM11 Magnetic Field
subplot(2,3,4);
imagesc(x*1000,y*1000,H_TM11);
axis xy;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
title('TM11 H-field');

% 5. Cutoff Frequency Comparison
subplot(2,3,5);
fc_GHz = [fc_TE10 fc_TM11]/1e9;
plot(1:2,fc_GHz,'-o','LineWidth',2,...
    'MarkerSize',6);
set(gca,'XTick',1:2,...
    'XTickLabel',{'TE10','TM11'});
xlim([0.8 2.2]);
ylabel('GHz');
title('Cutoff Frequency');
grid on;

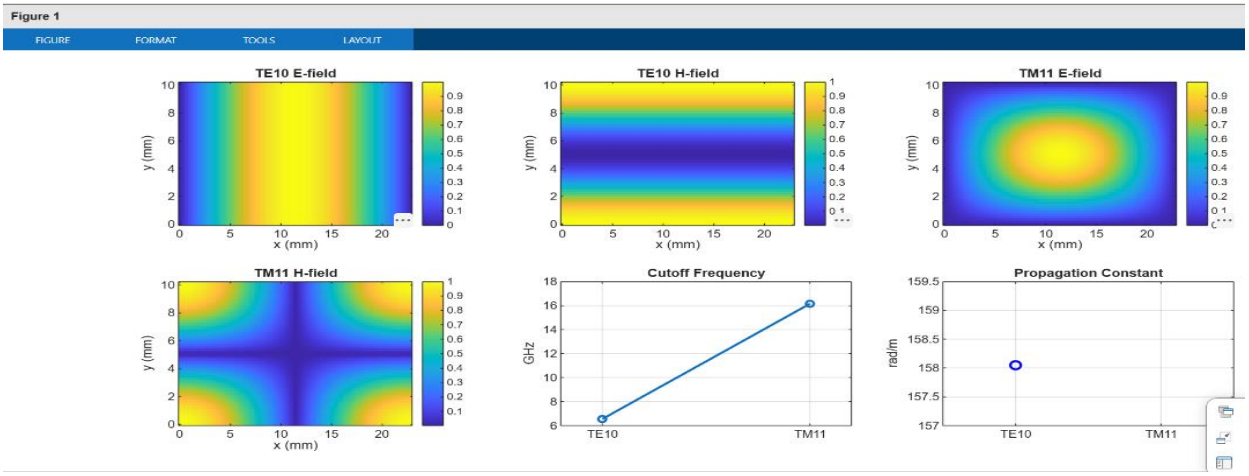
% 6. Propagation Constant
subplot(2,3,6);
plot(1,beta_TE10,'bo','MarkerSize',8,...
    'LineWidth',2);
hold on;
```

```
set(gca,'XTick',[1 2],...
    'XTickLabel',{'TE10','TM11'});
xlim([0.5 2.5]);
ylim([157 159.5]);
ylabel('rad/m');
title('Propagation Constant');
grid on;

% Display Results
fprintf('--- Microwave Mode Analysis ---\n');
fprintf('TE10 cutoff frequency = %.3f GHz\n',...
    fc_TE10/1e9);
fprintf('TM11 cutoff frequency = %.3f GHz\n',...
    fc_TM11/1e9);
fprintf('TE10 propagation constant = %.2f rad/m\n',...
    beta_TE10);

fprintf('\nAt 10 GHz:\n');
fprintf('TE10 propagates.\n');
fprintf('TM11 is below cutoff.\n');
```

5.Output:



6.Result:

The electromagnetic propagation modes were analyzed using MATLAB. The cutoff frequencies and propagation constant were calculated, and the electric- and magnetic-field distributions were visualized. The TE\(_{10}\) mode was found to propagate at 10 GHz because its cutoff frequency is approximately 6.562 GHz, whereas the TM\(_{11}\) mode remains below cutoff because its cutoff frequency is approximately 16.16 GHz.

